

Prompt-to-Production Debrief

Project: Student Report Card Portal

1. Which AI output misled you most?

The most misleading AI output occurred while debugging student report card access. Initial suggestions focused on report card retrieval queries and API endpoints. However, after detailed investigation, the actual issue was found to be the relationship between authenticated User records and Student records.

The AI helped identify possible areas to investigate, but manual debugging revealed that the mapping between User IDs and Student IDs needed correction. This experience demonstrated that AI suggestions should be treated as starting points rather than final answers.

2. How did you identify the issue?

The issue was identified through systematic debugging and verification.

The following steps were performed:

- Verified login functionality.
- Inspected authentication responses.
- Checked User and Student collections.
- Reviewed report card retrieval APIs.
- Compared User IDs and Student IDs.
- Tested report card access with different accounts.

By isolating each component, it became clear that the report card query was functioning correctly and the real issue was incorrect user-to-student mapping.

3. What did you verify manually?

Several important areas were manually verified throughout development.

Authentication & Authorization

- Login functionality for Admin, Faculty, and Student.
- Route protection.
- Role-based access restrictions.

Faculty Subject Assignment

- Faculty could edit only assigned subjects.

- Unauthorized updates were correctly rejected.

Student Report Cards

- Students could view only their own published report cards.
- Report card publishing workflow was validated.

Optimistic Concurrency

- Version conflicts were tested.
- Conflict handling workflow was verified.

Bulk CSV Import

- Validation rules were tested.
- Duplicate uploads were checked for idempotency.

Local Draft Recovery

- Draft saving and restoration were verified after page refresh.

Dashboard & Metrics

- KPI calculations were validated.
- Metrics endpoint responses were reviewed.

4. What would you improve if given another day?

If given additional development time, the following improvements would be implemented:

1. PDF report card generation.
2. Email notifications for published report cards.
3. Advanced analytics and reporting dashboards.
4. Improved automated testing coverage.
5. Enhanced audit reporting and monitoring.
6. Mobile-responsive UI improvements.
7. Additional data visualization features.

These enhancements would improve usability, maintainability, and overall user experience while preserving the current architecture.

Key Learning

AI tools significantly accelerated development, debugging, and documentation. However, the project reinforced the importance of verification, testing, and independent reasoning. Successful software development requires validating AI-generated suggestions rather than accepting them without evidence.